

Multiple Choice

1. **Answer:** B

Options: A) A procedure for testing libraries or other program components for vulnerabilities; B) Whole-system testing for security flaws and bugs; C) A security-minded form of unit testing that applies early in the development process; D) All of the above

Note: Correct option: B - Whole-system testing for security flaws and bugs

2. **Answer:** B

Options: A) Message Confidentiality; B) Message Integrity; C) Message Splashing; D) Message Sending

Note: Correct option: B - Message Integrity

3. **Answer:** A

Options: A) the matrix, if stored directly, is large and can be clumsy to manage; B) it is not capable of expressing complex protection requirements; C) deciding whether a process has access to a resource is undecidable; D) there is no way to express who has rights to change the access matrix itself

Note: Correct option: A - the matrix, if stored directly, is large and can be clumsy to manage

4. **Answer:** C

Options: A) crypter; B) virus; C) backdoor; D) key-logger

Note: Correct option: C - backdoor

5. **Answer:** B

Options: A) Installing and configuring a IDS that can read the IP header; B) Comparing the TTL values of the actual and spoofed addresses; C) Implementing a firewall to the network; D) Identify all TCP sessions that are initiated but does not complete successfully

Note: Correct option: B - Comparing the TTL values of the actual and spoofed addresses

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'a read outside bounds of a buffer'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Session layer'.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def even_num(x): | return True | return False`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def count_elim(num): | return count_elim`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key